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Synopsis Report

ON

"SMARTAQ: AN IOT-BASED AIR QUALITY MONITORING AN CONTROL SYSTEM FOR INDUSTRIAL AREA"

SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE

BACHELOR OF ENGINEERING
(COMPUTER ENGINEERING)

BY

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2025-2026

- **Title:**
  "SMARTAQ: AN IOT-BASED AIR QUALITY MONITORING AND CONTROL SYSTEM FOR INDUSTRIAL AREA"

- **Problem Definition:**

The SmartAQ system is designed to address critical challenges in air quality management within industrial environments, particularly in the chrome plating industry, where hazardous pollutants pose significant risks to worker health and environmental sustainability.

This section defines the specific problems the system aims to solve, outlines the project's objectives, identifies the research domains, and clarifies the scope to ensure a focused and achievable development process. Industrial settings, such as chrome plating facilities, generate a range of harmful air pollutants, including hexavalent chromium, volatile organic compounds (VOCs), ammonia(NH3), carbon monoxide (CO), nitrogen dioxide (NO2), methane (CH4), carbon dioxide (CO2), sulfur dioxide (SO2), ozone (O3), and particulate matter (PM2.5 and PM10). These pollutants are associated with severe health issues, such as respiratory disorders, cardiovascular diseases, and lung cancer, particularly for workers exposed to high concentrations over extended periods. Additionally, poor air quality contributes to environmental degradation, affecting ecosystems and regulatory compliance

- **Research Methodology:**

The SmartAQ system operates through a carefully designed methodology that integrates hardware, software, and analytical components to achieve real-time monitoring and forecasting. The following outlines the key elements and their roles: Sensors A suite of specialized sensors is deployed to capture comprehensive air quality data:

MQ-7 Sensor: Detects carbon monoxide (CO), a toxic gas prevalent in industrial emissions, with a detection range of 10 to 10,000 ppm. It provides critical data for assessing CO exposure risks.

MQ-135 Sensor: Monitors a broad spectrum of gases, including NH3, CO2, benzene (C6H6), an other pollutants, offering a holistic view of air quality conditions.

Dust Sensor: Measures particulate matter (PM2.5 and PM10), essential for tracking fine particles generated during chrome plating processes, which pose significant respiratory risks.

MQ-131 Sensor: Detects ozone (O3), a harmful gas produced in certain industrial activities, contributing to a detailed pollution profile.

Each sensor is connected to an Arduino Mega microcontroller, which collects: CO concentration data from the MQ-7. General air quality metrics from the MQ-135. PM2.5 and PM10 concentrations from the Dust Sensor. O3 levels from the MQ-131..

ESP32 Module

The ESP32 module serves as the wireless communication hub, receiving processed data from the Arduino Mega and transmitting it to a cloud-based IoT platform via Wi-Fi. Its dual-core processing and support for multiple protocols ensure reliable, real-time data transfer, enabling remote access for factory personnel.

IoT Platform

The IoT platform acts as the system's digital backbone, performing data storage, analysis, and visualization. It stores time-stamped sensor data in a database, runs AI models to predict pollution trends, and presents results through a user-friendly dashboard. This allows stakeholders to monitor air quality, identify patterns, and make informed decisions, such as activating ventilation systems when high pollutant levels are forecasted.

Data Flow

1. Sensors collect real-time data on pollutant concentrations and environmental parameters.
2. The Arduino Mega processes raw sensor data, converting it into usable formats (e.g.,ppm).
3. The ESP32 transmits data to the IoT platform using the MQTT protocol, ensuring secure and efficient communication.
4. The IoT platform stores data, applies AI models for forecasting, and visualizes results on a dashboard.
5. Predictive alerts trigger proactive interventions, such as ventilation activation, to mitigate pollution risks.

This methodology ensures a seamless integration of sensing, communication, and analytics, delivering a robust solution for industrial air quality management

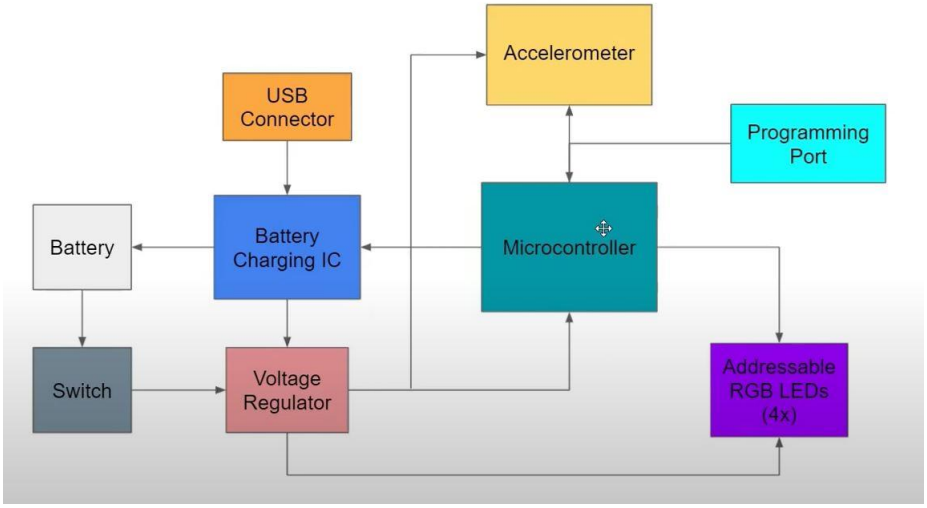


Fig: block diagram

- **Background:**

Air pollution remains a critical global challenge, with industrial operations significantly contributing to environmental degradation and public health risks. Factories and industrial processes release a complex mix of pollutants, including fine particulate matter (PM2.5 and PM10), sulfur dioxide (SO2), nitrogen dioxide (NO2), carbon monoxide (CO), volatile organic compounds (VOCs), and heavy metals. These contaminants deteriorate air quality, leading to serious health conditions such as asthma, heart disease, and lung cancer, particularly for workers exposed to elevated pollutant levels in industrial settings. Beyond human health, air pollution disrupts ecosystems, reduces agricultural yields, and fuels climate change, ozone depletion, and acid rain. For example, regions with heavy industrial activity, such as parts of Eastern Europe, report PM2.5 levels far exceeding safe thresholds, resulting in higher mortality rates and reduced life expectancy. In mining and manufacturing hubs, emissions from machinery and processes further intensify air quality issues, endangering both workers and nearby communities. Mitigating these impacts requires robust monitoring and swift action to control pollution levels, ensuring safer environments and sustainable industrial practices.

- **Proposed System:**

The chrome plating industry generates hazardous pollutants, including hexavalent chromium, volatile organic compounds (VOCs), ammonia (NH3), carbon monoxide (CO), nitrogen dioxide (NO2), methane (CH4), carbon dioxide (CO2), sulfur dioxide (SO2), ozone (O3), and particulate matter (PM2.5 and PM10). These pollutants pose significant health risks to workers, ranging from respiratory issues to long-term conditions like lung cancer, and contribute to environmental degradation. Traditional monitoring methods, reliant on periodic sampling or stationary stations, lack real-time capabilities and predictive insights, making them inadequate for addressing dynamic pollution challenges in industrial environments.

SmartAQ overcomes these shortcomings by deploying a network of IoT sensors to continuously monitor air quality and transmitting data to a cloud platform for analysis. AI models, including Long Short-Term Memory (LSTM), Random Forest, and Linear Regression, process this data to forecast pollution levels, enabling proactive measures. The system's ability to predict high pollution events hours in advance allows factory managers to implement interventions, such as enhancing ventilation or adjusting processes, to maintain safe air quality. This holistic approach not only improves worker health and safety but also supports regulatory compliance and sustainable industrial practices, setting a new standard for environmental

• Tools and Technologies Used

The SmartAQ system relies on a combination of hardware, software, and development tools to achieve its functionality. The following tools and technologies are utilized:

1. **Programming Languages**:
    - **C++**: Used for programming the Arduino Mega microcontroller, leveraging libraries for sensor integration and communication.
    - **Python**: Employed for cloud-based data analysis and AI model development, including LSTM, Random Forest, and Linear Regression.
2. **Hardware**:
    - **Sensors**: MQ-7 (CO), MQ-135 (NH3, CO2, benzene), MQ-131 (O3), and Dust Sensor (PM2.5, PM10) for air quality monitoring.
    - **Arduino Mega 2560**: The primary microcontroller for sensor data collection and processing.
    - **ESP32 Module**: Provides Wi-Fi connectivity for data transmission to the cloud.
3. **Integrated Development Environment (IDE)**:
    - **Arduino IDE**: An open-source platform for writing, compiling, and uploading C++ code to the Arduino Mega, supporting sensor and communication libraries.
4. **Design and Documentation Tools**:
    - **StarUML**: Used to create Unified Modeling Language (UML) diagrams, such as use case, class, and sequence diagrams, to visualize system architecture and interactions.
    - **LaTeX**: Employed for professional documentation and report generation.
    - **PDF Tools**: For compiling and sharing project deliverables in a standardized format.

- ## Expected Outcomes:

The implementation of SmartAQ yielded impressive results, validating the effectiveness of IoT and AI in air quality management. Key findings include:

- **Comprehensive Monitoring**: The system accurately measured a wide range of pollutants (CO, NH3, NO2, CO2, O3, PM2.5, PM10) and environmental parameters (temperature, humidity), providing a holistic view of air quality.
- **Sensor Performance**: Calibration ensured high sensitivity and reliability, with the MQ-7 detecting CO levels as low as 10 ppm and the Dust Sensor accurately tracking PM concentrations.
- **Data Integrity**: The IoT architecture facilitated robust data collection, transmission, and storage, with no significant data loss during testing.
- **Predictive Accuracy**: AI models achieved prediction accuracies above 80%, with LSTM excelling in time-series forecasting and Random Forest performing well for PM2.5 predictions.
- **Operational Impact**: Real-time alerts and predictive insights enabled timely interventions, reducing exposure risks and improving air quality in the test facility.

These results underscore the system's potential to address air pollution challenges in industrial environments, providing actionable data for stakeholders

- ## Expected Path for Your Research:

1. Literature Review:
A comprehensive literature review is fundamental to any research project, serving as a critical step to contextualize the study, identify existing challenges, and uncover opportunities for innovation. By analyzing prior work in the field, researchers can pinpoint gaps, refine methodologies, and build upon established knowledge to address unresolved issues. This review focuses on the evolution of air quality monitoring, the shortcomings of traditional approaches, the transformative impact of Internet of Things (IoT) and Artificial Intelligence (AI) technologies, their practical applications, and the specific challenges faced in industrial environments like the chrome plating industry. It concludes by highlighting a critical research gap in real-time, factory-based air quality monitoring and forecasting, which this study aims to address.
Reviewing relevant academic papers, journals, and books.

Summarizing key findings, methodologies, and challenges identified in the literature.

Identifying gaps and areas for innovation in current research.

2. Problem Definition and Scope Refinement:

The SRS covers the hardware and software components required for the SmartAQ system, including IoT sensors, microcontrollers, wireless communication modules, cloud-based data processing, and user interfaces. It specifies the system's functionality for monitoring air pollutants, transmitting data wirelessly, analyzing trends, forecasting pollution levels, and proposing control mechanisms. The document does not address the implementation of fully automated control systems or the monitoring of all possible pollutants, focusing instead on the specified pollutants and initial control framework design.

3. Methodology Design:

1 System Operation

The SmartAQ system operates through a distributed network of IoT sensors strategically placed across the chrome plating facility. These sensors continuously monitor a comprehensive set of pollutants and environmental parameters, including CO, NH3, NO2, CH4, CO2, SO2, O3, VOCs, formaldehyde (CH2O), hydrogen sulfide (H2S), oxygen (O2), particulate matter (PM1.0, PM2.5, PM10), temperature, and humidity. The system's methodology is designed to ensure accurate data collection, real-time processing, and predictive analytics, enabling proactive air quality management.

Components and Workflow

1. **Air Sampling Mechanism**:
    - An air sampling pump draws ambient air into the sensing node for analysis.
    - A fan ensures continuous airflow, expelling measured air and preparing the system for new samples.
2. **Sensor Network**:
    - Sensors measure multiple pollutants and environmental factors, providing a detailed air quality profile.
    - Data are transmitted to the Arduino Mega for initial processing and formatting.
3. **Data Transmission**:
    - The ESP32 module sends processed data to a cloud-based IoT platform via Wi-Fi, using the MQTT protocol for secure and efficient communication.
4. **Data Processing and Analysis**:
    - The IoT platform stores data in a database and applies AI models to analyze trends and forecast pollution levels.
    - Machine learning algorithms (LSTM, Random Forest, Linear Regression) identify patterns and predict high pollution events.
5. **Visualization and Alerts**:
    - A dashboard displays real-time data, historical trends, and predictive alerts, enabling factory personnel to monitor air quality and make informed decisions.

Alerts are triggered when pollutant levels approach hazardous thresholds, prompting actions like ventilation activation.

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